

Integrating Jira and Bitbucket

1. In Jira, click on the 'Settings' button; after that, click on the product integration button.
2. Next, click on 'DVCS accounts'.
3. Then click on 'Link Bitbucket Cloud Account'.
4. Click on 'Grant access'.
5. Enable 'Automatically sync all repositories' and 'Smart Commits' for all repositories. Click on 'Okay'.
6. Go to your particular newly created issue, and you will see there is an option in the development section called '+ create branch'. Click on that option.
7. In this screen, you can select 'Repository'. Select Type (Bugfix, feature, release, other) in 'Branch', and then give the new branch a name. Click on the 'Create' button.
8. In that issue, you will see your branch.

Integrating Bitbucket and Jenkins

Take the following steps to install the Jenkins plugin.

1. Go to the Jenkins dashboard: *Manage Jenkins -> Manage Plugins -> Available Plugins*.
2. Search for Bitbucket.
3. Select 'Bitbucket Build Status Notifier' plugins and click on 'Install'.
4. Restart Jenkins.
To create an OAuth consumer, you first need to get an OAuth consumer key/secret from Bitbucket.
 1. Log in to your Bitbucket account.
 2. Click your account name and then click on 'Settings' from the menu bar.
 3. Click 'OAuth' from the menu bar.
 4. Press the 'Add consumer' button.
 5. The system requests the following information:
 - a. Give a representative name to the consumer, e.g., Jenkins build status notifier.
 - b. Although it is not used, a callback URL must be set, e.g., <http://>

The screenshot shows the Jenkins interface with a sidebar on the left containing links like 'Up', 'Status', 'Configure', 'Scan Multibranch Pipeline Now', 'Scan Multibranch Pipeline Log', 'View as plain text', 'Multibranch Pipeline Events', 'Delete Multibranch Pipeline', 'People', 'Build History', 'Project Relationship', 'Check File Fingerprint', 'Rename', 'Pipeline Syntax', and 'Credentials'. The main area displays the 'Scan Multibranch Pipeline Log' for a branch named 'DEMO-1-add-feature-1'. The log text includes: 'Started by user Akshay_Ithape', '[Sun Oct 06 09:22:56 UTC 2019] Starting branch indexing...', 'using GIT_ASKPASS to set credentials bitbucket1', 'Setting origin to https://akshayithape1252@bitbucket.org/akshayithape1252', 'Fetching & pruning origin...', 'Listing remote references...', 'Checking branch master', 'Checking branch DEMO-1-add-feature-1', 'Changes detected: DEMO-1-add-feature-1 (5e30e03fa402a2457f996730237c)', 'Scheduled build for branch: DEMO-1-add-feature-1', 'Processed 2 branches', '[Sun Oct 06 09:22:57 UTC 2019] Finished branch indexing. Indexing to', and 'Finished: SUCCESS'.

Figure 8: Jenkins pipeline logs

The screenshot shows the Bitbucket 'Branches' page for a repository named 'Demo'. The page has a sidebar with 'Source', 'Commits', 'Branches', and 'Pull requests'. The main area shows a table of branches with columns for 'Branch', 'Behind', 'Ahead', and 'Updated'. The 'DEMO-1' branch is listed as the active branch, updated 3 hours ago.

Figure 9: Build status in Bitbucket

The screenshot shows the Jira 'Open issues' page for a project named 'DEMO-1'. The issue 'Add feature 1' is selected. The page shows the issue title, description, activity, and comments. The issue is in the 'Add feature 1' state.

Figure 10: Build status in Jira

The screenshot shows the Jira 'Builds' page for a project named 'Development DEMO-1'. The page shows a table of builds with columns for 'Pipeline', 'Latest build', and 'Test Results'. The build '#2' is listed as the latest build, with a green checkmark indicating a successful build.

Figure 11: Build status in Jira with build number